Md Mehedi Hasan

PhD Student, Electrical and Computer Engineering, University of Maine, Orono, ME

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Research Interests

Efficient large language model inference, robotics and multi-robot SLAM, 3D vision and neural rendering, and AI on edge devices.

Education

PhD, Electrical and Computer Engineering — In progress

University of Maine, Orono, ME. Advisor: Vikas Dhiman

BSc, Computer Science and Engineering — 2020 – 2023

Bangladesh University, Dhaka. Ranked 1st in class, CGPA 3.91 / 4.00

Publications

[1] **M. M. Hasan**, B. R. Biswas, X. Hou, and Y. Guan, "ForestRAG-AR: An AR and RAG framework for context-aware silviculture assistance," conference paper, Dec. 2025.

[2] M. T. R. Sobur, **M. M. Hasan**, M. R. Khatun, M. M. Mia, W. Islam, M. S. Iqbal, and M. M. Hossain, "Machine learning-enhanced SPR-based optical biosensor for cancer cell detection," *Optics & Laser Technology*, vol. 192, p. 113796, Elsevier, 2025.

Research Projects

Speculative Decoding Across CPU and GPU — University of Maine

- Split speculative decoding so a small draft model runs on the CPU while the large model verifies on the GPU.
- Benchmarked GPT-2 / GPT-2 Large, DistilGPT-2 / GPT-2, Llama 3.2 1B / 3B, and MiniLLM-212M / Llama 3.2 1B.
- Profiled latency per transformer block with PyTorch forward hooks, CUDA events, and CPU perf counters.

Multi-Robot SLAM on Edge Devices — University of Maine

- Built a split ORB-SLAM3 system across two Jetson Orin robots with RealSense L515 depth cameras and a shared server over ZeroMQ.
- Implemented map merging with duplicate map point removal and live inter-robot distance tracking.

Animatable 3D Humans from Video — University of Maine

- Replacing the body estimator in NeuMan with 4DHumans (HMR 2.0); evaluating with PSNR, SSIM, and LPIPS on six benchmark sequences.
- Wrote a qualifying-exam paper on animatable 3D avatars using SMPL body models and NeRF-style rendering.

Solar-Powered LoRa Mesh Sensing — University of Maine

- Designed a low-power sensor network: STM32 nodes with LoRa and flash storage reporting to a Jetson base station with LTE uplink.

Experience

Graduate Researcher, University of Maine — Present

Research in LLM inference, robotics, and 3D vision with Vikas Dhiman.

Machine Learning Engineer, Inovetix — Sep 2023 – Nov 2024

- Built machine learning models to predict customer satisfaction and guide business decisions.
- Developed interactive dashboards to turn large datasets into actionable insights.

Research Assistant, Bangladesh University Dec 2022 – Dec 2024

- Developed machine learning frameworks for healthcare applications and PCF-SPR biosensor design.

Outreach

Live Object Detection Demo for High School Students University of Maine

Built a hands-on open-vocabulary detection demo (YOLO-World) on a Jetson Orin for visiting students.

Skills

Languages	Python, C++, C, Java, JavaScript
Machine learning	PyTorch, Hugging Face Transformers, TensorFlow, Keras, scikit-learn
Robotics and vision	ORB-SLAM3, ROS, OpenCV, PyTorch3D, Intel RealSense
Systems	Docker, CUDA, NVIDIA Jetson, STM32, ESP32, LoRa, Linux

Training

Introduction to Java Programming (Coursera); C for Everyone: Programming Fundamentals; Web Development, MERN stack (Programming Hero).

References available on request.